Maternal and child health

Goal: Understand the concept of facility and community based maternal and child health care

Specific objectives:

- Outline the goal and focus of MCH and family planning
- Describe MCH and family planning services
- Describe community base approach to MCT and FP

Introduction

- Maternal health is the health of women during pregnancy, childbirth, and the postpartum period.
- It encompasses the health care dimensions of family planning, preconception, prenatal, and postnatal care in order to reduce maternal morbidity and mortality.
- Preconception care can include:
    - Education,
    - Health promotion,
    - Screening
    - and other interventions among women of reproductive age to reduce risk factors that might affect future pregnancies.

- Postnatal care issues include:
    - Recovery from childbirth,
    - Concerns about newborn care,
    - Nutrition,
    - Breastfeeding,
    - Family planning

- The goal of prenatal care is to:
    - Detect any potential complications of pregnancy early,
    - Prevent them if possible,
    - Direct the woman to appropriate specialist medical services as appropriate.

- Primary areas of focus in MCH include:
    - Increasing healthy birth outcomes;
    - Promoting and assuring comprehensive primary care for children, including children with special health care needs;
    - Promoting healthy lifestyles among school-age youth, ages 6-21, including children with special health care needs;
    - Assuring access to safe, healthy child care up to age six, including children with special health care needs;
    - Preconception, pregnancy testing and planning.

- In Africa, one out of 210 mothers dies during pregnancy or delivery.
- Most maternal deaths are due to causes directly related to:
    - Pregnancy
    - Childbirth,
    - Unsafe abortion
    - Obstetric complications such as:

        - Severe bleeding
        - Infection
        - Hypertensive disorders
        - Obstructed labor
    - Other indirect cause like:
        - Relatively low rate of institutional deliveries,
        - Due to transport problems
        - Lack of infrastructure,
        - Due to cultural prejudices and resistance against giving birth outside the family circle

- The fifth Millennium Development Goal (MDG) aims to reduce the maternal mortality ratio by 75% between 1990 and 2015.
- In Kenya, maternal mortality remains high at 488 maternal deaths per 100,000 live births.
- The MCH/FP clinic is founded on quality and afford-able services that advocate for safe motherhood, proper child development, and a healthy family.

MCH Services

- The services offered in MCH clinic are as follows:
    - Focused antenatal care.
    - Targeted post natal care.
    - Providing all family planning methods: short term, long term, permanent i.e. vasectomy & tubal ligation.
    - Screening of Breast and Cervical cancer in all women of reproductive ages.
    - Giving immunization i.e. BCG, Pentavalent, Polio, Measles.
    - Growth monitoring for children under the age of five.
    - Counseling and testing of antenatal mothers and partners on PMTCT - eMTC
    - Create awareness through Micro teaching on various topics i.e. Nutrition, Breast feeding, Weaning etc.

Immunization

- The basic principle of immunization is to administer into a healthy person a vaccine that will prevent that person from getting a certain disease.
- Vaccines may be of:
    - Live attenuated: (e.g. Rubella, OPV, Measles and BCG),
    - Inactivated or killed (e.g. Hib, IPV)
    - Micro-organisms
    - Detoxified toxins (e.g. Tetanus)
- Generally, several vaccines can be given at the same time. This is important since you do not know when you will see the child again.
- BCG, OPV, DPT-HeB-Hib and Measles vaccines can be given simultaneously if the child is of the appropriate age and has not received the early immunizations.
- A critically ill child needing hospital admission must be given the appropriate vaccines upon recovery
- Remember:
    - That a slight fever and/or other minor illness should not prevent you from immunizing a child
    - To inform mothers/child-caretakers about possible side effects of each of the given

vaccines
- To record all vaccinations on tally sheets and on the Child-Health immunization cards and instruct the mothers always to bring the cards along with them when taking children to a health facility
- To instruct mothers to return the child for the next immunization date as indicated on the card
- That vaccines are easily destroyed by heat and rendered ineffective
- To handle the disposal of used sharp syringes appropriately
- To ensure appropriate cold storage of the vaccines and follow the recommended cold- chain instructions for each of the vaccines carefully.

Facts about the common vaccine-preventable diseases

Whooping cough (Pertussis):

- A communicable disease.
- Spread by droplets.
- Symptoms include:
  - Severe cough followed by a whoop and vomiting, Leads to malnutrition, Can cause death, Severe in under 1 year old.

Diphtheria:

- An infectious disease.
- Spread by droplets.
- Symptoms include: Difficulty in breathing, Swallowing, Enlarged neck.
- Very severe when it occurs.

Tetanus: [see tetanus]:

- A clinical syndrome involving primarily the central nervous system and resulting from the tetanus toxins.
- Enters through open wounds, cuts and umbilical stump.
- Symptoms include: Stiffness, Locked jaw, Inability to suckle and muscle spasms.
- Has a very high mortality (>50%).
- Immunizing pregnant mothers ensures protection of her new born baby.

Measles (Rubeola):

- Highly infectious killer disease.
- Symptoms include: Rash, Fever, Cough, Red eyes;
- Is associated with: blindness, malnutrition, deafness, pneumonia and death.

Poliomyelitis (Infantile Paralysis):

- An acute communicable disease.
- Spread by droplets and oro-faecal contamination.
- Symptoms include:
  - Pain and flaccid paralysis in limbs, fever, vomiting; can lead to permanent deformity, and can cause death.

Tuberculosis:

- A communicable disease.
- Spread by droplet.
- Symptoms include: fever, wasting, deep chesty cough, Night sweats, May have

lymphadenopathy.
- Leads to lowered resistance to other diseases and may be fatal.

Hepatitis B:
- Is a highly infectious disease.
- Transmitted mainly by:
  - Parenteral route,
  - From person to person
  - By close contact
  - Through exchange of body fluids such as:
    - Saliva, Secretions from open wounds, Blood, Vaginal secretions and semen.
- Transmission between children is common, since they are often more infectious than adults.
- Transmission from carrier mothers can occur in up to 80% of babies during the perinatal period,
- Infection may be transmitted either:
  - Vertically (transplacentally from mother to unborn baby)
  - Horizontally by close contact.

Childhood immunization schedule in Kenya

Vaccine	Age	Remarks
BCG POLIO (OPV O)	At birth Birth Dose	Or at first contact with child
DPT_1-HeB_1-Hib_1 DOSE POLIO (OPV 1)	6 weeks (1 ½ months)	Or at first contact with child after that age
DPT_2-HeB_2-Hib_2 DOSE POLIO (OPV 2)	10 weeks (2 ½ months)	4 weeks after DPT 1 and OPV 1 can also be given anytime after this period, when in contact with the child.
DPT_3-HeB_3-Hib_3 DOSE POLIO (OPV 3)	14 weeks (3 ½	4 weeks after DPT 2 and OPV 2 can also be given any time after this period, when in contact with the child.

	months)	
Measles	9 months	May be given between 6 and 9 months if child is admitted to hospital for any other illness. Repeat at 9 months as per KEPI schedule.

Immunization schedule for pregnant mothers (TT2+)

Tetanus Toxoid (TT)	Pregnant women	1st dose with first pregnancy or subsequent pregnancy
		2nd dose - 4 wks after first dose
		3rd dose - 6 months after 2nd dose
		4th dose - at least 1 year after the third dose
		5th dose - at least 1 year after the fourth dose

Immunizing a pregnant mother ensures protection of her newborn baby.

Vitamin A supplementation schedule

Dosage (in U)	When to give
<6 months - 50,000	First dose at 9 months then every 6 months (twice per year) up to the age of 60 months.
6-12 months- 100,000 iu	
>12 months 200,000 iu	

VACCINE DOSAGE AND ROUTE OF ADMINISTRATION

VACCINE DOSE (*=Live vaccines)	ROUTE OF ADMINISTRATION
BCG* Child under 1 year, 0.05 ml. Child over 1 year, full dose, 0.1 ml	Intra-dermally into upper outer part of left forearm, at the junction of the upper and middle thirds. If given correctly a small wheal appears at the site of the injection. Inform the mother that a small sore will appear in 2-6 weeks. Let this heal by itself. It will leave a small scar. If no reaction develops, the vaccination should be repeated after 3 months.
POLIO (OPV)* 2 drops by mouth. Follow manufacturer's instructions on dosage	Read the instructions on the bottle. Give vaccine by mouth. Use dropper provided. If child spits or vomits repeat the dose.
Pentavalent vaccine consisting **of DPT-HeB-Hib** 0.5 ml	Intramuscularly in the upper outer part of the thigh.
HB 0.5 ml - child 1.0 ml - adult	Intramuscularly in the upper outer part of the thigh for child and deltoid (left) for adult
Measles* 0.5 ml	Subcutaneously or intramuscularly in upper outer part of the arm (deltoid muscle)
Tetanus Toxoid (TT) 0.5 ml.	Intramuscularly in the outer part of the upper-arm (deltoid muscle)

Notes: Hands should be washed before and after handling vaccines. All the vaccines and diluents must be kept cold. DPT, HB, and TT vaccines are damaged if kept below 0°C and therefore should never be frozen. Always check Vaccine Vial Monitor (VVM).

Contra-indications. A definite severe reaction to a preceding vaccine dose is a contraindication to further doses.
Live vaccines should not be given to individuals with impaired immune response e.g. leukaemia, HIV/AIDS.

The intramuscular route should not be used in patients with bleeding disorders such as haemophilia or thrombocytopaenia.

HIV/AIDS INFECTION AND IMMUNIZATION
Asymptomatic children infected with HIV should receive all standard Kenya Expanded Programme on Immunization (KEPI) vaccines. Non-immunized children with symptomatic HIV infection should receive all standard vaccines except BCG.

Side-effects and adverse reactions to vaccinations
Range from mild to severe for various vaccines.

BCG Vaccine
These include injection abscess, regional or wide-spread lymphadenitis, osteomyelitis and disseminated BCG infection. These should be treated with anti-tuberculosis drugs **[see 12.8. TB]**

Oral Polio Vaccine
Adverse reactions rarely occur.

Measles Vaccine
Adverse reactions include fever, mild rash and rarely convulsions, encephalitis and sub-acute schlerosing pan-encephalitis (SSPE).

DPT (Diphtheria, Pertussis, Tetanus)
Most adverse reactions are attributed to the pertussis component. Minor reactions include, pain at the site of injection and fever. Major reactions are persistent crying, high pitched cry, excessive somnolescence, convulsions, encephalopathy and coma.

Recombinant DNA Hepatitis B Vaccine
Side effects include pain, fever and swelling at the site of injection.

Family Planning and Contraceptive methods, types and Options in Kenya

- The choice of one method of contraception against the other depend on:
  - individual medical condition
  - taste of an individual or partner,
- The most common method of family planning is hormonal method except condom followed by mechanical methods. The most reliable method is surgical method and the most unreliable method is natural methods.
- Type of family planning and contraceptives can largely be divided into four categories
  - Natural
  - Barrier

- Hormonal
- Surgical

NATURAL METHODS OF FAMILY PLANNING AND CONTRACEPTIVES METHODS IN KENYA

- It is called natural because it does not involve use of drug or device
- The following are the available method or types of natural family planning methods
- The following make up natural family planning methods:
  - **Withdrawal Method (coitus interrupters)**
    - Warning this method can only be used in highly motivated couples.
    - It is useful as a temporary method for example supplies for of regular method are not available or the couples are together for only a short time.
  - **Safe period (rhythm method)**
    - This one means that couples avoid sex the time the woman eggs (ovum) are ready for fertilization.
    - This is around mid-month of a woman's circle.
    - This means that there should no sex for the couples who are using this method between day 8(eight) and day 20 (twenty) of a woman's circle i.e. a total of 12 days without sex in a month.
    - Have you ever seen a cow on heat? Unfortunately this is the period at which women folks are on heat so this method like the withdrawal method require highly motivated couples
  - **Breastfeeding amenorrhea method**
    - Risk of pregnancy in exclusively breastfeeding couples in the first 6 (six) months so long as the woman has not experienced menses is 2 % (percent)
    - It is a reliable method so long as the above conditions are intact.
  - **Cervical mucus membrane**
    - This is a slightly technical method and couples are advised against using this if they don't understand the physiology of ovulation.
- **Basal body temperature (BBT) method**
  - Like cervical mucus membrane this is also a slightly cumbersome method because it involve daily recording of a chart.
  - Check the temperature five minutes before waking up and enter in ascertain chart after woman's eggs are released (ovulation) temperature rises by (0.2 – 0.6) C intercourse should be avoided when the temperature is low and resumed when the temperature has risen up next period.
  - like you might have noticed this method is very unreliable form of family planning, what if one gets sick an the temperature rises, what if one fails to record the temperature.
  - This method can be easily manipulated by women folk to deny men folks there conjugal rights by just manipulating the temperatures.

Note: Natural family planning requires high motivation and so makes them not particularly reliable method of family planning.

BARRIER METHODS OF CONTRACEPTION AND FAMILY PLANNING IN KENYA

- These method or type of family planning include:
  - Male and Female Condoms
  - Spermicides
  - Diaphragms
  - Cervical Caps
- Whereas condoms, diaphragms, and cervical caps are mechanical barriers, spermicides are chemicals that interfere with the movement of the sperm and its ability to fertilise the egg.
- Currently in Kenya, the use of diaphragms, cervical caps, and spermicides is negligible.
- In addition, scientific evidence has shown that repeated and high-dose use of the spermicide might cause vaginal and cervical irritation or abrasions, which could increase the risk of infection with HIV.

HORMONAL METHODS OF FAMILY PLANNING AND CONTRACEPTIVE METHODS IN KENYA

- This is the most commonly used method or type of family planning and contraceptive after condom
- They include hormone taken orally, by injection or implanted to prevent development and maturation of woman's eggs (ovum's)
- Common hormonal FP and contraceptives methods are:
  - Emergency contraception (e-pills)
  - Ordinary pills
  - Injectable contraceptives
  - Contraceptive Implants (J adelle, Implanon, Zarin)
  - Hormone-releasing intrauterine systems
    - This method involves use of devices which may sometime have some form of drug (hormone) attached to it ,
    - Their mode of function is by placing a barrier so that sperm does not get into the fallopian tube to fertilize the woman's egg.
    - The following are the available IUCD type of family planning and contraceptives methods in Kenya
      - Copper-based IUCDs
      - Hormone-releasing devices

VOLUNTARY SURGICAL CONTRACEPTION

- Apart from use of condom there is another area in which men can contribute to family planning directly.
- The following are the available surgical family planning methods:
  - Tubal legation
  - Vasectomy

Community-Based Maternal, Newborn and Child Health Services

- Aims to reduce the number of deaths among mothers, newborns, and young children by improving the health services available in four rural districts
- Provides women with greater access to good-quality, reliable health services.

- It helps to ensure that district health facilities offer essential health services for women and children;
- It provides training for health care workers in emergency care for pregnant women, new mothers, and newborn babies;
- It strengthens the management and supervision practices in health care facilities.
- It improves the ability of men and women in the communities to recognize, prevent, and respond to maternal and child health issues.
- The community MCH services include:
    - Organizing community events and mass media campaigns,
    - Training community theatre groups,
    - Selecting peer youth educators and male champions to raise awareness about potential health issues;
    - Conducting home visits to provide pre-natal and post-natal care and care for newborns and young children;
    - Equipping community health workers to provide services in remote areas and to refer patients to health centres where necessary.
- It emphases the role that men can play in helping to improve the health of women and children.

Growth Monitoring

- This is the monitoring of the growth of a child through nutrition.
- Weight of the child is monitored using the CWC card
- The weight of the child is monitored for 5 years